

PROJECT PROPOSAL & FEASIBILITY REPORT

Project Title: IndoorPulse: Infrastructure-Free Autonomous Indoor Positioning and Emergency Routing System

Course Code: UCS503 (Software Engineering Project)

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1. EXECUTIVE SUMMARY

Indoor navigation remains a major unresolved challenge in ubiquitous computing due to the severe attenuation and multipath degradation of Global Navigation Satellite System (GNSS/GPS) signals inside enclosed structures. Existing commercial indoor positioning systems (IPS) rely heavily on specialized, static hardware infrastructure—such as Bluetooth Low Energy (BLE) beacon grids, Ultra-Wideband (UWB) anchors, or dense Wi-Fi fingerprint databases. These systems impose unsustainable installation, maintenance, and calibration overhead while failing completely during facility power disruptions.

IndoorPulse delivers an autonomous, offline-first Indoor Positioning System built natively for Android. The application couples smartphone-integrated Inertial Measurement Unit (IMU) sensor fusion with a 2D Monte Carlo Particle Filter and a localized topological graph engine. This report evaluates the technical, operational, and economic feasibility of deploying IndoorPulse in multi-corridor institutional environments.

2. PROBLEM STATEMENT & PROJECT SCOPE

The Problem

Modern architectural facilities (such as university academic blocks, healthcare complexes, and transit hubs) feature expansive, multi-corridor layouts where physical wayfinding induces cognitive fatigue and navigational delay. In time-critical scenarios such as emergency evacuations or critical medical response, disorientation presents severe life-safety risks.

Target Users

- Students and faculty navigating complex university infrastructure.

- Visitors and patients in healthcare facilities.
- First responders requiring immediate, reliable orientation in unmapped or smoke-compromised environments without power.

System Scope

- **In Scope:** Continuous client-side step detection, dynamic stride length estimation, inertial heading translation, localized topological mapping (nodes and edges), Dijkstra-driven shortest-path routing, and decoupled deliverable verification via a self-hosted cloud portal.
- **Out of Scope (Current Milestone):** Multi-building campus-wide macro-routing, automated 3D LiDAR point-cloud ingestion, and active continuous cloud telemetry streaming.

3. FEASIBILITY ANALYSIS MATRIX

Feasibility Dimension	Assessment Criteria	Technical Reality in IndoorPulse	Verdict
Technical	Real-time sensor polling and processing latency	IMU polling via Android SensorManager at 50 Hz. Dual-threshold Schmitt Trigger executes in $O(1)$ time. 400-particle Monte Carlo filter bounded by topological spatial masks.	Feasible
Economic	Hardware acquisition, deployment, and operational costs	\$0 Hardware Footprint. Operates entirely on consumer smartphone sensors. Cloud architecture runs within AWS EC2 free-tier allocation, Backblaze B2 (10 GB free tier), and GitHub Pages.	Feasible
Operational	User adoption friction, setup	Graph creation handled once via an administrative "Surveyor Mode". Zero external network connection required	Feasible

Feasibility Dimension	Assessment Criteria	Technical Reality in IndoorPulse	Verdict
	overhead, offline resilience	for end-user navigation; fully offline Room/SQLite storage.	
Legal & Compliance	User privacy, data governance, institutional guidelines	Zero personal data or user telemetry collected. Adheres strictly to UCS503 non-blackbox architecture guidelines.	Feasible

4. TECHNICAL METHODOLOGY & ARCHITECTURE

1. Kinematic Dead Reckoning & Digital Filtering

Consumer-grade Micro-Electro-Mechanical Systems (MEMS) accelerometers suffer from thermal noise and baseline drift. Unconstrained double integration of raw acceleration produces quadratic position error growth within seconds. IndoorPulse resolves this through discrete gait cycle segmentation:

- **Dual-Threshold Schmitt Trigger:** Employs hysteresis thresholds ($T_{\text{high}} = 1.25 \text{ m/s}^2$, $T_{\text{low}} = -0.65 \text{ m/s}^2$) with a minimum temporal lockout of 280 ms. This guarantees the complete suppression of hand tremors and stationary desk vibrations while reliably capturing true heel-strike impacts.
- **Dynamic Stride Scaling:** Implements a biomechanical bounce-frequency formulation:

$$\text{Stride Length} = k \times (a_{\text{max}} - a_{\text{min}})^{(0.25)} \times (f_{\text{stride}})^{\alpha}$$

where k is a calibrated individual scalar and f_{stride} is the instantaneous step frequency, keeping stride error margins within 5%.

2. Heading Translation & Spatial Transformation

Raw magnetometer data suffers from severe magnetic anomaly distortions near indoor reinforced structural steel. IndoorPulse utilizes Android's fused `TYPE_ROTATION_VECTOR` (integrating gyroscopic rates with gravity vectors) and maps the physical body frame to digital map coordinate space using a fixed Euclidean translation:

- $X_{(t+1)} = X_t + \text{Stride_Length} \times \cos(\theta_{\text{fused}} - \theta_{\text{declination}})$
- $Y_{(t+1)} = Y_t + \text{Stride_Length} \times \sin(\theta_{\text{fused}} - \theta_{\text{declination}})$

3. Topological Constraint (Particle Filtering)

To prevent unconstrained lateral drift through structural walls, a localized 2D Monte Carlo filter initializes particles along valid graph segments. When a step event triggers, candidate particles are projected forward. Any particle that intersects non-walkable wall geometry or diverges from the topological edge is eliminated and resampled from the highest-probability distribution cluster.

5. RISK ASSESSMENT & MITIGATION STRATEGY

Risk Description	Impact	Likelihood	Mitigation Strategy
Cumulative IMU Drift over extended corridors	High	Medium	Apply Zero-Velocity Updates (ZUPT) when sensor variance drops below stationary thresholds, coupled with hard edge-snapping along known topological segments.
Hardware sensor variance across smartphone vendors	Medium	High	Implement dynamic adaptive thresholding where the Schmitt Trigger baseline dynamically tracks the running mean of acceleration peaks over a 5-step rolling window.
Multi-floor transition ambiguity	High	Low	Implement explicit QR-anchor scans or manual floor selector toggles at vertical transition stairwells and elevators for non-barometric devices.
Cloud presentation portal failure	Medium	Low	Maintain automated static asset fallbacks on GitHub Pages that load local cached documents if the AWS EC2 compute endpoint is unreachable.

6. WORK BREAKDOWN & TIMELINE

Phase	Milestone / Deliverables	Timeline	Status
Phase 1: Architecture Setup	Problem definition, 3-tier web portal deployment (AWS EC2, Backblaze B2, GitHub Pages), Android project repository initialization.	Weeks 1–3	Complete
Phase 2: Core Algorithm Development	Schmitt Trigger gait state machine, biomechanical stride estimation, Jetpack Compose reactive MapCanvas coordinate transforms.	Weeks 4–7	Complete
Phase 3: Integration & Testing	Dijkstra shortest-path pathfinding, physical 50m corridor testing, mathematical edge-weight calibration.	Weeks 8–10	In Progress
Phase 4: Advanced Upgrades & Viva	A* pathfinding optimization, Wi-Fi landmark drift resets, final technical report compilation, prototype viva defense.	Weeks 11–14	Scheduled

7. PROJECT RECOMMENDATION

The feasibility assessment confirms that **IndoorPulse is practically, computationally, and economically viable**. By abandoning active external beacons in favor of smartphone-native sensor fusion and topological graph algorithms, the system eliminates capital deployment costs while guaranteeing continuous functionality in high-stress and offline operating conditions. The technical foundation has been validated through functional prototypes, and development should proceed directly into the integration and spatial-calibration testing phase.